ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE C

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70): magnetism and induction, resistor networks, transformers and DC machines, resistivity and temperature, transistors, using a multimeter, logic gates and care of test equipment. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Pure silicon to which a small amount of phosphorus has been added becomes …

A an insulator.
B an N-type semiconductor.
C a P-type semiconductor.
D a superconductor.

1.2 The unit of magnetic flux is the …

A weber.
B tesla.
C henry.
D farad.

1.3 Which instrument shows the shape of a changing voltage on a screen?

A Ammeter
B Ohmmeter
C Oscilloscope
D Wattmeter

1.4 The output of a two-input AND gate is 1 only when …

A any input is 1.
B both inputs are 0.
C one input is 0.
D both inputs are 1.

1.5 Fleming's right-hand rule is used to find the direction of the …

A induced current in a generator.
B force on a motor conductor.
C current in a resistor.
D charge on a capacitor.

1.6 A secondary cell …

A cannot be recharged.
B can be recharged.
C produces alternating current.
D contains no electrolyte.

1.7 The resistance of a conductor increases when its …

A length decreases.
B cross-sectional area increases.
C length increases.
D material is changed to silver.

1.8 Which component stores energy in a magnetic field?

A Resistor
B Inductor
C Capacitor
D Diode

1.9 A Zener diode is mainly used for …

A voltage regulation.
B amplification.
C storing charge.
D measuring light.

1.10 Which ONE of the following is good practice when putting away a digital multimeter?

A Leave it switched on.
B Store it in a damp place.
C Leave flat batteries inside it.
D Switch it off and store it in its case.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 In a parallel circuit, the voltage across each branch is the same.

2.2 A multimeter on the resistance (ohms) range may safely be used on a live circuit.

2.3 Lenz's law states that an induced current opposes the change that caused it.

2.4 A step-up transformer has fewer turns on its secondary winding than on its primary winding.

2.5 Holes are the majority charge carriers in P-type material.

2.6 A NOT gate has two inputs.

2.7 The resistance of copper increases as its temperature rises.

2.8 An LED must be connected the right way round (forward biased) to give light.

2.9 Capacitors connected in series have a larger total capacitance than any one of them.

2.10 A PLC output can switch a contactor that starts a motor.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		**COLUMN B**
3.1	Majority charge carriers in N-type material	A	reluctance
3.2	Region at a PN junction with no free charge carriers	B	commutator
3.3	Gate whose output is 1 only when its two inputs are different	C	electrons
3.4	Tool used to remove the insulation from the end of a wire	D	cut-off
3.5	Joining components and wires with molten metal	E	holes
3.6	Part of a DC machine that reverses the current in the armature coils	F	depletion layer
3.7	State of a transistor that is fully switched off	G	frequency
3.8	Opposition of a magnetic circuit to magnetic flux	H	wire stripper
3.9	Number of complete cycles in one second	I	EX-OR
3.10	Board with copper strips, used to build permanent circuits	J	Veroboard
		K	soldering
		L	amplitude
		M	NAND

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 Define the following concepts:

4.1.1 Magnetic flux (1)

4.1.2 Magnetic flux density (1)

4.1.3 Permeability (1)

4.1.4 Electromagnet (1)

4.2 A coil of 400 turns carries a current of 2,5 A. The length of its magnetic path is 0,2 m. Calculate the:

4.2.1 magnetomotive force (mmf) (2)

4.2.2 magnetic field strength (H) (2)

4.3 State Faraday's TWO laws of electromagnetic induction. (4)

4.4 A conductor 0,3 m long moves at 5 m/s at right angles through a magnetic field with a flux density of 0,8 T. Calculate the emf induced in the conductor. (3)

[15]

QUESTION 5

5.1 Three resistors of 10 Ω, 15 Ω and 30 Ω are connected in parallel. This group is connected in series with a 5 Ω resistor across a 24 V DC supply. Calculate the:

5.1.1 resistance of the parallel group (3)

5.1.2 total resistance of the circuit (1)

5.1.3 total current (2)

5.1.4 voltage across the 5 Ω resistor (2)

5.2 An ideal transformer has 1 150 turns on its primary winding and is connected to 230 V. It must give 12 V on its secondary winding.

5.2.1 Calculate the number of turns needed on the secondary winding. (3)

5.2.2 The secondary winding supplies 2 A. Calculate the primary current. (2)

5.3 Name FOUR main parts of a DC motor. (4)

5.4 Differentiate between a DC series motor and a DC shunt motor. (2)

5.5 Give ONE application of a DC series motor. (1)

[20]

QUESTION 6

6.1 Calculate the resistance of an aluminium conductor that is 500 m long and has a cross-sectional area of 2,5 mm². Take the resistivity of aluminium as 0,028 μΩ·m. (4)

6.2 A copper coil has a resistance of 20 Ω at 0 °C. Calculate its resistance at 60 °C if the temperature coefficient of resistance of copper at 0 °C is 0,00427 /°C. (3)

6.3 Draw the symbol of an NPN transistor and label its THREE terminals. (3)

6.4 Explain how a transistor is used as a switch so that a 5 V Arduino pin can switch on a 12 V relay. (3)

6.5 Write down the relationship between the emitter, collector and base currents of a transistor. (2)

[15]

QUESTION 7

7.1 Explain, step by step, how to measure the current through a resistor in a circuit with a digital multimeter, safely. (6)

7.2 Draw the symbol and the truth table of a TWO-input NAND gate. (5)

7.3 Draw the logic diagram for the Boolean expression $X = A \cdot B + \overline{C}$. The diagram must have three inputs, A, B and C. (5)

7.4 Give FOUR ways to look after electrical testing equipment. (4)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Ohm's law: $V = I \times R$	Power: $P = V \times I = I^2 \times R$
Series: $R_T = R_1 + R_2 + R_3$	Parallel: $1 \div R_T = 1 \div R_1 + 1 \div R_2 + 1 \div R_3$
Resistivity: $R = \rho \times L \div A$	Temperature: $R_t = R_0(1 + \alpha_0 \times t)$
Magnetomotive force: $mmf = N \times I$	Field strength: $H = mmf \div l$
Flux density: $B = \Phi \div A$	Induced emf: $e = B \times l \times v$
Transformer: $V_p \div V_s = N_p \div N_s = I_s \div I_p$	Efficiency: $\eta = P_{out} \div P_{in} \times 100\%$
Transistor: $I_E = I_C + I_B$	Frequency: $f = 1 \div T$
Capacitors in parallel: $C_T = C_1 + C_2 + C_3$	Capacitors in series: $1 \div C_T = 1 \div C_1 + 1 \div C_2 + 1 \div C_3$
Battery capacity: $Ah = I \times t$	Kirchhoff: ΣI in = ΣI out; Σ voltage rises = Σ voltage drops